

SARVESH PANDEY

Data Science | Machine Learning | NLP

Lucknow, India | 7704022244 | psarvesh991@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

Computer Science undergraduate specializing in Data Science and Machine Learning, with hands-on experience building and deploying end-to-end ML systems. Skilled in Python, SQL, Scikit-learn, PyTorch, NLP, model evaluation, probability calibration, FastAPI, and Streamlit. Developed production-oriented projects in credit risk classification and multi-label emotion intelligence with emphasis on model validation, threshold optimization, REST APIs, and cloud deployment.

Education

B.Tech in Computer Science & Engineering

Dr. A.P.J. Abdul Kalam Technical University, Lucknow

Class XII – CBSE: 72% (2022) Class X – CBSE: 82% (2020)

2022 – 2026

CGPA: 8.03/10

Technical Skills

Languages	Python, C++, SQL, MySQL
Data Science	Pandas, NumPy, Scikit-learn, Statistics, EDA, Feature Engineering
Machine Learning	Regression, Classification, Clustering, Ensemble Methods
Deep Learning	TensorFlow, Keras, PyTorch, CNN, RNN, LSTM, Transformers
Visualization	Matplotlib, Seaborn, Power BI
Tools / Backend	Git, GitHub, VS Code, Jupyter, Streamlit, FastAPI
Core CS	DBMS, Operating Systems, OOPs

Projects

Credit Risk Analysis

[GitHub](#) | [Live App](#)

Python, Scikit-learn, XGBoost, LightGBM, FastAPI

- Built an end-to-end credit risk analysis system using 20K+ loan records and 11 borrower and loan risk features to generate probability-based loan risk assessments.
- Benchmarked Logistic Regression, XGBoost, and LightGBM models and deployed LightGBM as the final predictive model.
- Applied probability calibration and a tuned decision threshold of 0.3516 for probability-based risk classification and decision making.
- Achieved approximately 0.94 accuracy, 0.93 precision, 0.77 recall, and 0.84 F1-score on the evaluation set.
- Productionized real-time inference using FastAPI and Pydantic with an interactive frontend for loan risk prediction.

Emotion Intelligent System

[GitHub](#) | [Live App](#)

Python, PyTorch, NLP, DistilBERT, FastAPI, Streamlit, Docker

- Deployed an end-to-end multi-label NLP classification system on the GoEmotions dataset, predicting 28 emotion categories across 54K+ samples.
- Performed data cleaning, leakage detection, EDA, class-imbalance analysis, tokenization, and preprocessing.
- Benchmarked TF-IDF + Logistic Regression, BiLSTM, and DistilBERT models with weighted loss and validation-based threshold optimization; achieved 0.5205 Macro F1, 0.5714 Micro F1, and 0.5699 Weighted F1 on the test set.
- Productionized real-time inference using FastAPI, Pydantic, Streamlit, Docker/Docker Compose, Git/GitHub, Render, and Streamlit Community Cloud.
- Implemented model comparison, per-class evaluation, threshold optimization, and error analysis.

Certifications & Achievements

Python (Basic) – HackerRank

[Certificate](#)

SQL (Basic) – HackerRank

[Certificate](#)

SQL (Intermediate) – HackerRank

[Certificate](#)

5-Star Gold Badge in SQL – HackerRank

[Profile](#)

200+ DSA Problems – TakeUForward

[Profile](#)